

Relay Attacks in Intra-handshake Attestation and Proposal

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(ACK: Eric Rescorla, Juho Forsén, Markus Rudy, Mariam Moustafa,
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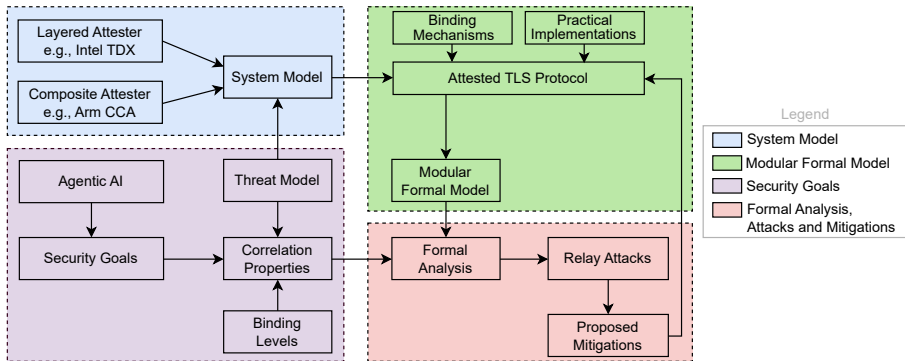
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Quick Recap of Approach



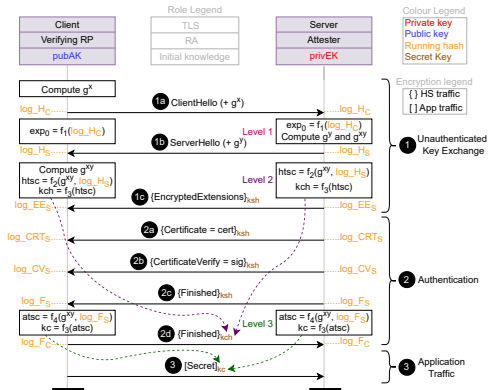
Potential Mechanisms for Binding (Server as Attester)

S. No.	Binding material	Value of rdata field
1	Client's TLS nonce	nc
2	Client's attestation nonce	na
3	Early exporter	exp_0
4	Server's public key	pubEK
5	Client's attestation nonce and early exporter	na exp_0
6	Client's attestation nonce and server's public key	na pubEK
7	Nonce, server's public key, and early exporter	na pubEK exp_0

- Meta AI: #1
- Edgeless Systems Contrast and Cocos AI: #6
- draft-fossati-tls-attestation-06: #5
- Discussion: attested-tls-poc¹: #6?
- Discussion: Any other useful binding material/combination within SEAT WG charter?

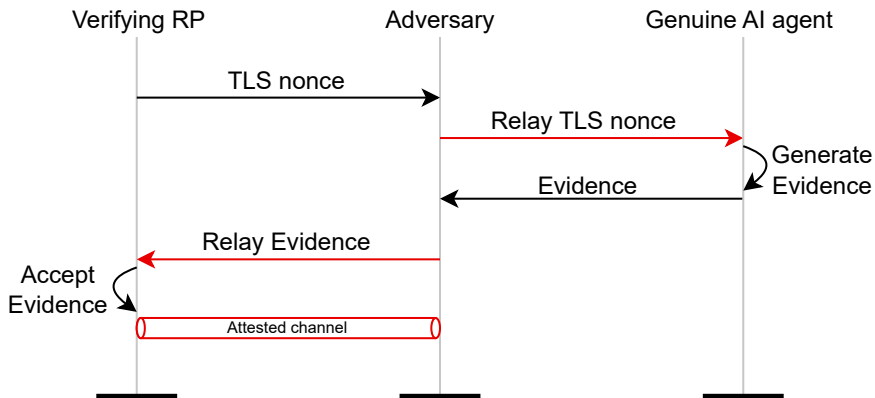
¹<https://github.com/CCC-Attestation/attested-tls-poc>

Strong Binding vs. Relay of Evidence (Abstracted)

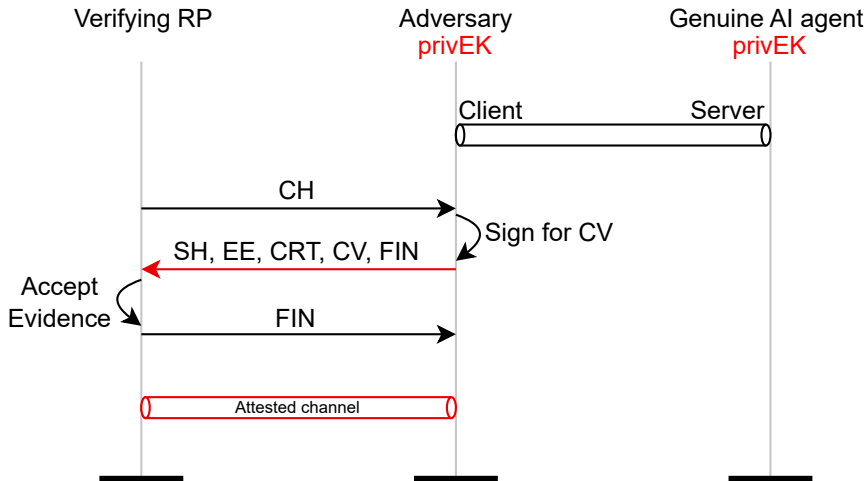


- htsc: used for encryption of clientFinished message (2d).
 - Irrelevant for security goals
 - Server **not yet authenticated** at this point
- atsc: used for encryption of application data (client's secret, e.g., decryption key)
 - Relevant for security goals

Relay Attack 1 (Abstracted): Mechanism # 1 (nc)



Relay Attack 2 (Abstracted): Mechanism # 4 (pubEK)



Need for Implementation Guidance

- Meta's AI: TLS nonce (hence, no Evidence freshness)
- Conveyance of attestation nonce
 - Edgeless Systems Contrast (previously) and Cocos AI abuse SNI (intended for hostname)
 - Edgeless Systems Contrast (now) abuse ALPN (intended for protocol names)
- CCC Attested TLS proof-of-concept²: No update for 3 years
- **Discussion:** Any other intra-handshake attestation implementation within SEAT WG charter?

²<https://github.com/CCC-Attestation/attested-tls-poc>

Proposal: **Post**-handshake Attestation³

- Spec (non-binding): draft-fossati-seat-expat
- Detailed rationale: draft-usama-seat-intra-vs-post
- **Real** implementors interest with **real** use cases
 - Edgeless Systems (Markus Rudy)
 - Zama (Ayoub Benaissa)
 - Flashbots (Peg Jones)
 - GA4GH and ELIXIR Europe (Pavel Nikonorov)
 - You?

³<https://github.com/ccc-attestation/governance/issues/20>